

NUCLEI

Important formulas and concepts directly from the NCERT-

- The positive charge and mass are densely concentrated at the centre of the atom forming its nucleus.
- The nucleus contains most of the mass of an atom.

❖ Atomic masses and composition of nucleus-

- **Atomic mass unit (u)**, defined as $1/12^{\text{th}}$ of mass of carbon(C^{12}) atom.

$$1\text{u} = 1.67 \times 10^{-27} \text{ kg}$$

- **The positive charge in the nucleus is that of protons.** The photon is stable.
- Isotopes of an element have same atomic no. but different mass no.
- Unlike proton, neutron is unstable and it decays into a proton, an electron and an antineutrino and has a mean life of about **1000s**. **it is however stable inside nucleus.**
- **Z - atomic no. = no. of protons**
N – neutron no. = no. of neutrons
A – mass no. = Z+N = total no. of protons and neutrons = no. of nucleons
- Nuclear species are shown by ${}_Z\text{X}^A$.

➤ **As atoms of isotopes have identical electronic structure they have identical chemical behaviour and placed in same location in periodic table.**

➤ **Isobars have same mass no. but have different atomic no.**

➤ **Isotones have same no. of neutrons but different atomic no.**

❖ **Size of the nucleus-**

➤ Nucleus of mass no. A has radius,

$$R = R_0 A^{1/3}$$

Where, $R_0 = 1.2 \times 10^{-15} \text{m}$.

➤ volume of nucleus which is proportional to R^3 is proportional to A.

➤ **density of nucleus is independent of A , for all nuclei.**

Its value is approximately $2.3 \times 10^{17} \text{kg/m}^3$

❖ **mass-energy relation-**

➤ **Einstein showed that mass is another form of energy and one can convert mass-energy into other forms of energy.**

➤ **Einstein's mass-energy equivalence relation,**

$$E = mc^2$$

❖ **Nuclear binding energy-**

➤ It is expected that mass of nucleus is equal to total mass of its individual protons and neutrons. However, the nuclear mass M is found to be always less than this.

➤ The difference in mass of a nucleus and its constituents , ΔM is

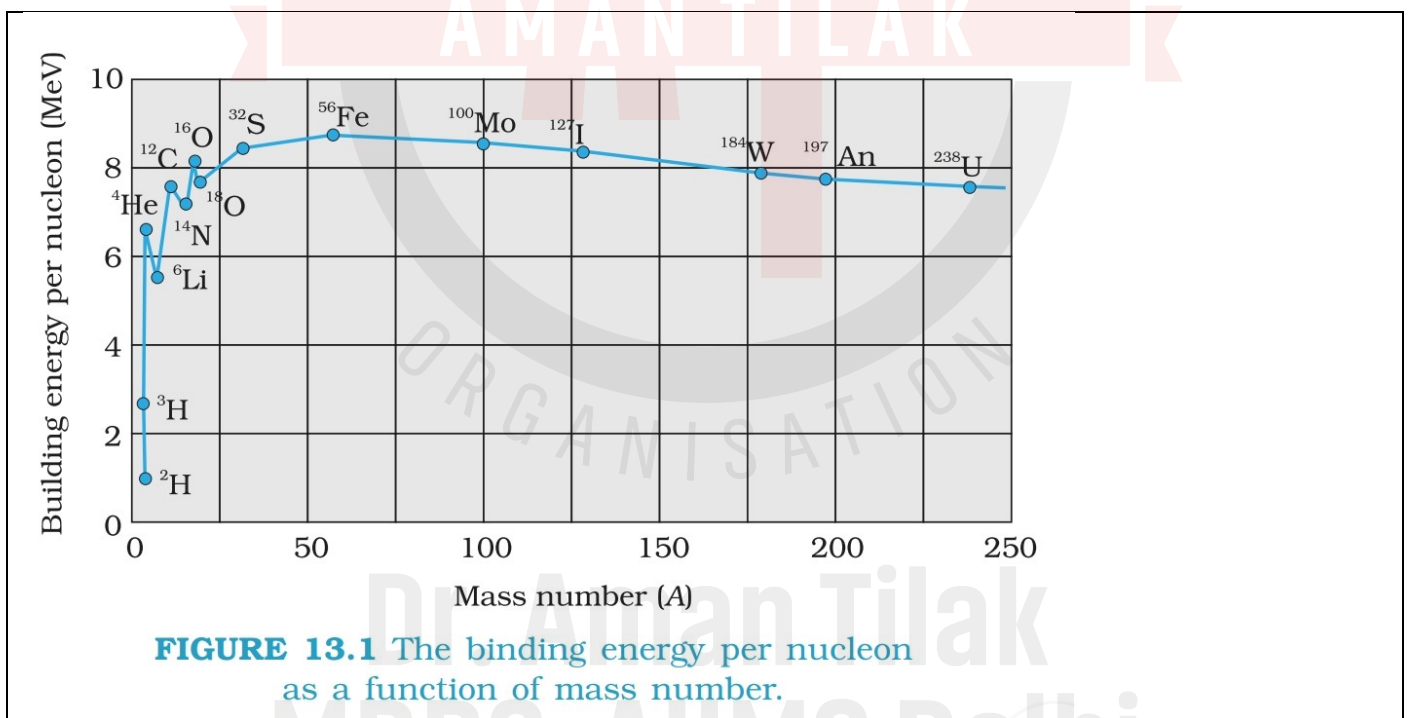
$$\Delta M = [Zm_p + (A-Z)m_n] - M$$

➤ If one wants to break nucleus into neutrons and protons, some extra energy has to be supplied. Or if neutrons and protons are brought together to form a nucleus, some energy is released, **this energy is known as ‘binding energy of nucleus’** and given by,

$$E_b = \Delta Mc^2$$

➤ Binding energy per nucleon,

$$E_{bn} = E_b/A$$



➤ The main features of plot-

- 1) E_{bn} is independent of atomic no. for nuclei of middle mass no. ($30 < A < 170$). The curve has a maximum of about 8.75 MeV for $A = 56$ and has a value of 7.6 MeV for $A = 238$.

2) E_{bn} is lower for both light nuclei ($A < 30$) and heavy nuclei. ($A > 170$)

➤ For average mass nuclei, binding energy per nucleon is approx. 8 MeV.

❖ Nuclear Force-

➤ There must be a strong attractive force which can overcome repulsion b/w protons and can bind protons and neutrons into tiny nuclear volume, this force is known as 'Nuclear Force'.

➤ Properties of nuclear force-

1) It is much stronger than coulomb force b/w protons and gravitational force.

2) This force is applicable only to a short range, that's why known as 'short range force'.

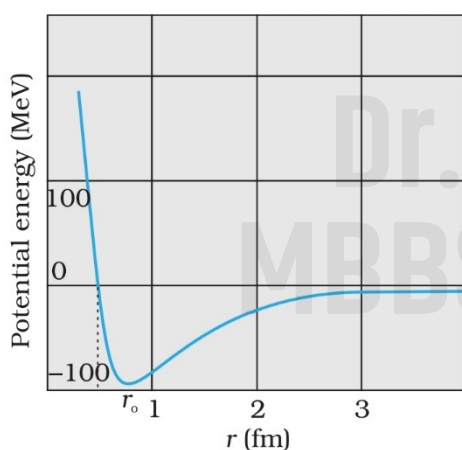


FIGURE 13.2 Potential energy of a pair of nucleons as a function of their separation. For a separation greater than r_0 , the force is attractive and for separations less than r_0 , the force is strongly repulsive.

- The potential energy is minimum at distance r_0 of about 0.8 fm.
Force is attractive for distance greater than 0.8 fm
Force is repulsive for distance smaller than 0.8 fm

3) Nuclear force doesn't depend on charge.

It is same b/w neutron-neutron, proton-neutron and proton-proton.

❖ Nuclear energy-

- a heavy nucleus ($A > 170$) with lower E_{bn} , if breaks into two smaller nuclei of higher E_{bn} , nucleons get more tightly bound. This implies that energy is released in this process (**fission**).
- If two lighter nuclei ($A < 30$) with lower E_{bn} , joins to form a heavy nucleus with higher E_{bn} , energy is again released in this process(**fusion**).
- **Greater the binding energy, less is the total mass of a nucleus.**
That's why if a nucleus transforms into a nucleus of greater binding energy, there is a mass defect which is converted into energy and thus released.

❖ Fission –

- An example of fission is ,

SUMMARY

1. An atom has a nucleus. The nucleus is positively charged. The radius of the nucleus is smaller than the radius of an atom by a factor of 10^4 . More than 99.9% mass of the atom is concentrated in the nucleus.
2. On the atomic scale, mass is measured in atomic mass units (u). By definition, 1 atomic mass unit (1u) is $1/12^{\text{th}}$ mass of one atom of ^{12}C ; $1\text{u} = 1.660563 \times 10^{-27} \text{ kg}$.
3. A nucleus contains a neutral particle called neutron. Its mass is almost the same as that of proton
4. The atomic number Z is the number of protons in the atomic nucleus of an element. The mass number A is the total number of protons and neutrons in the atomic nucleus; $A = Z + N$; Here N denotes the number of neutrons in the nucleus.

A nuclear species or a nuclide is represented as ${}_Z^A\text{X}$, where X is the chemical symbol of the species.

Nuclides with the same atomic number Z , but different neutron number N are called *isotopes*. Nuclides with the same A are *isobars* and those with the same N are *isotones*.

Most elements are mixtures of two or more isotopes. The atomic mass of an element is a weighted average of the masses of its isotopes and calculated in accordance to the relative abundances of the isotopes.

5. A nucleus can be considered to be spherical in shape and assigned a radius. Electron scattering experiments allow determination of the nuclear radius; it is found that radii of nuclei fit the formula

$$R = R_0 A^{1/3},$$

where $R_0 = \text{a constant} = 1.2 \text{ fm}$. This implies that the nuclear density is independent of A . It is of the order of 10^{17} kg/m^3 .

6. Neutrons and protons are bound in a nucleus by the short-range strong nuclear force. The nuclear force does not distinguish between neutron and proton.

7. The nuclear mass M is always less than the total mass, Σm , of its constituents. The difference in mass of a nucleus and its constituents is called the *mass defect*,

$$\Delta M = (Z m_p + (A - Z)m_n) - M$$

Using Einstein's mass energy relation, we express this mass difference in terms of energy as

$$\Delta E_b = \Delta M c^2$$

The energy ΔE_b represents the *binding energy* of the nucleus. In the mass number range $A = 30$ to 170 , the binding energy per nucleon is nearly constant, about 8 MeV/nucleon .

8. Energies associated with nuclear processes are about a million times larger than chemical process.
9. The Q -value of a nuclear process is

$$Q = \text{final kinetic energy} - \text{initial kinetic energy.}$$

Due to conservation of mass-energy, this is also,

$$Q = (\text{sum of initial masses} - \text{sum of final masses})c^2$$

10. Radioactivity is the phenomenon in which nuclei of a given species transform by giving out α or β or γ rays; α -rays are helium nuclei; β -rays are electrons. γ -rays are electromagnetic radiation of wavelengths shorter than X-rays.
11. Energy is released when less tightly bound nuclei are transmuted into more tightly bound nuclei. In fission, a heavy nucleus like ${}^{235}_{92}\text{U}$ breaks into two smaller fragments, e.g., ${}^{235}_{92}\text{U} + {}^1_0\text{n} \rightarrow {}^{133}_{51}\text{Sb} + {}^{99}_{41}\text{Nb} + 4 {}^1_0\text{n}$
12. In fusion, lighter nuclei combine to form a larger nucleus. Fusion of hydrogen nuclei into helium nuclei is the source of energy of all stars including our sun.

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